

VISGRAF · IMPA · OCTOBER 2026

Active Foveal 3-D Vision

A fixate–reconstruct–saccade playground for 360° scene reconstruction

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THE IDEA

High resolution only where you look, and a whole room assembled by looking around

A small fovea

Each look measures a 12° patch in metric 3-D, not the whole field.

Chosen saccades

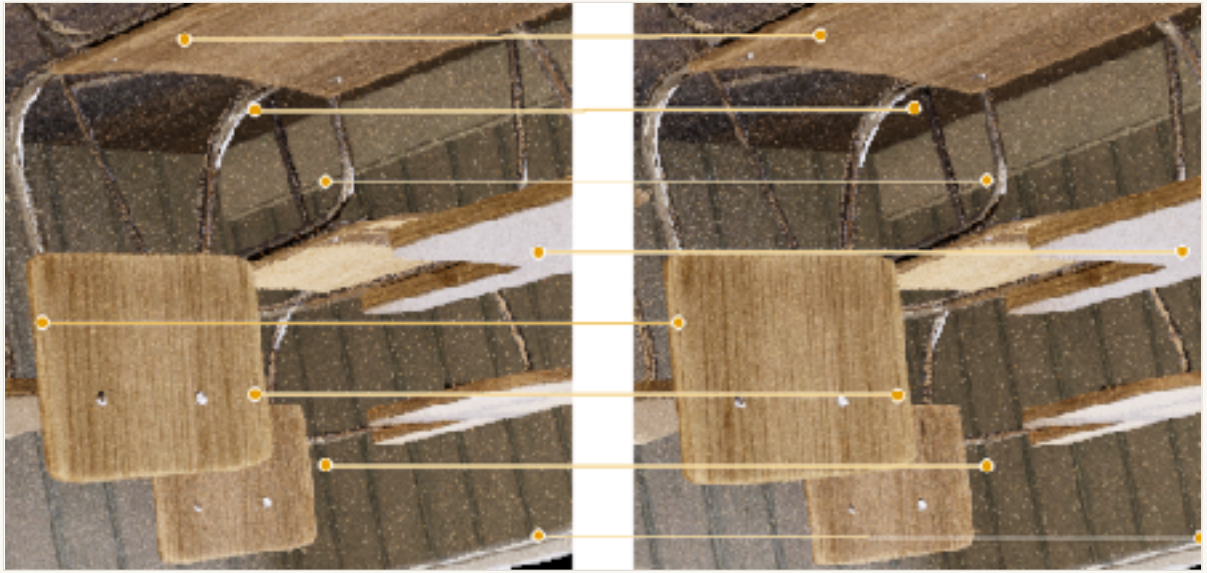
The next gaze is picked from what has not been seen yet.

Lasting memory

Every measurement lands in one persistent map of the room.

THE OBSERVER

A binocular observer with a fixed head



Fixation #179: the left and right foveal cores. Corresponding points share a row (yellow).

Part	Setting
Head	fixed; checked every fixation
Eyes	63 mm baseline
Vergence	2.1 m along the gaze
Fovea	12° core, 256 × 256 px
Stereo	PERFECT / ORACLE correspondence
Scene	Blender Classroom, 64 spp

Source: technical report “Active Foveal 3-D Vision”, sections 4–5

THE LOOP

Fixate, reconstruct, remember, saccade

STEP 1

Fixate

Render the stereo pair at the chosen gaze.

STEP 2

Reconstruct

Match every foveal pixel; triangulate about 65 k points.

STEP 3

Remember

Fuse into one 12-mm surfel map; mark the 1° sphere.

STEP 4

Saccade

Step to a neighbour, or jump to unseen space.

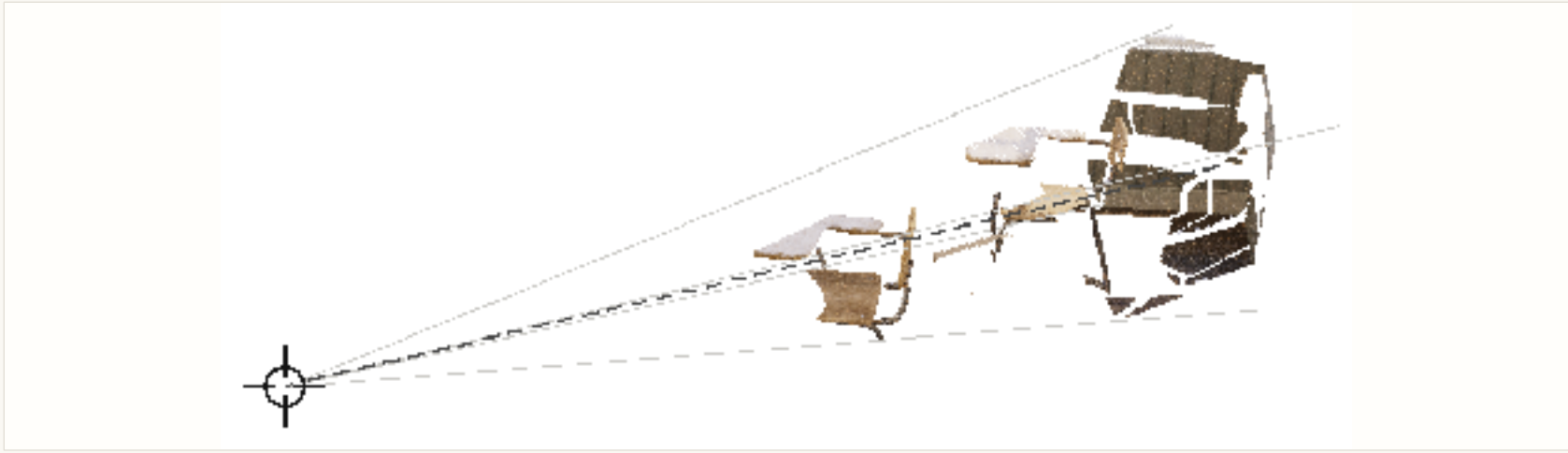
repeat until no unseen view remains

ONE FIXATION

From one gaze to about 65,000 metric points



Range from the left eye, 0.5–5.5 m



The same points in the head frame H0: head (crosshair), gaze ray (dashed), 12° core (grey lines)

60,778
metric points from fixation #179

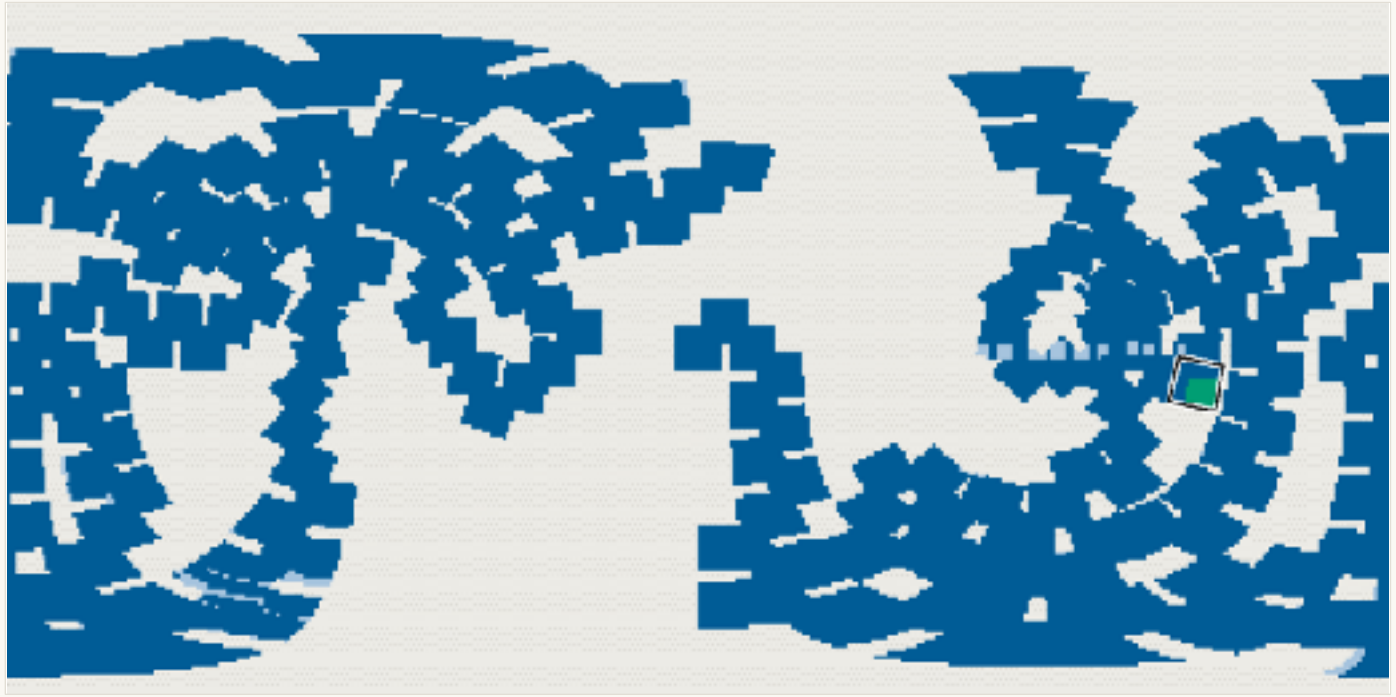
36,481
fused into existing surfels

24,297
new surfels added to the map

MEASURED · official baseline run, fixation #179; median over all fixations: 65,150 points of 65,536 rays

CYCLOPEAN MEMORY

The controller remembers a 1° sphere



After fixation #179 (360° × 180°). Green: cells this fixation added; outline: its 12° footprint.

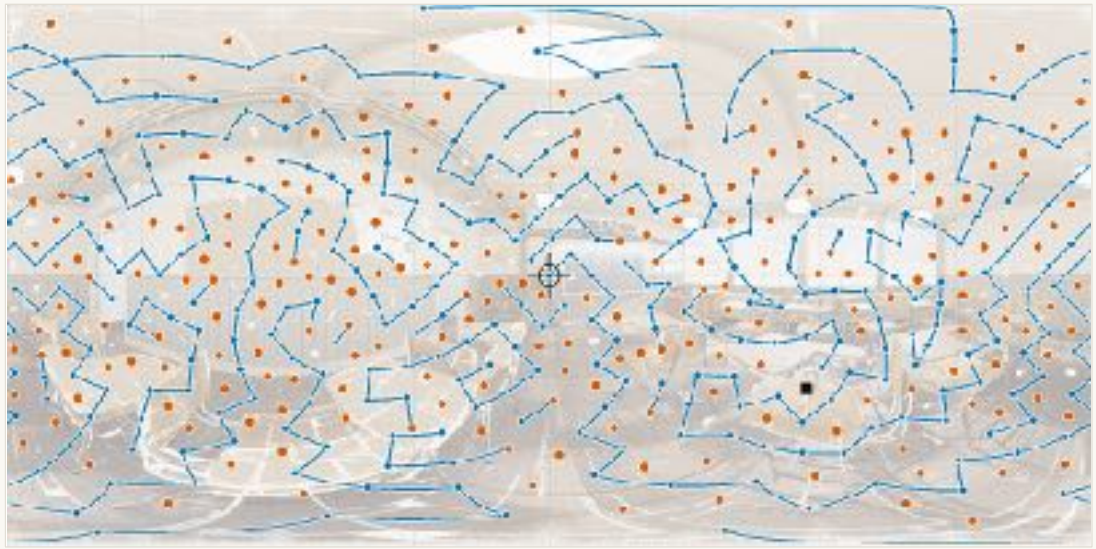
- **DEPTH**: seen and reconstructed
- **SEEN**: looked at, no depth
- **UNSEEN**: not yet looked at

Gaze is chosen from this sphere and the current fixation alone. The 3-D map is memory and output, never a control input.

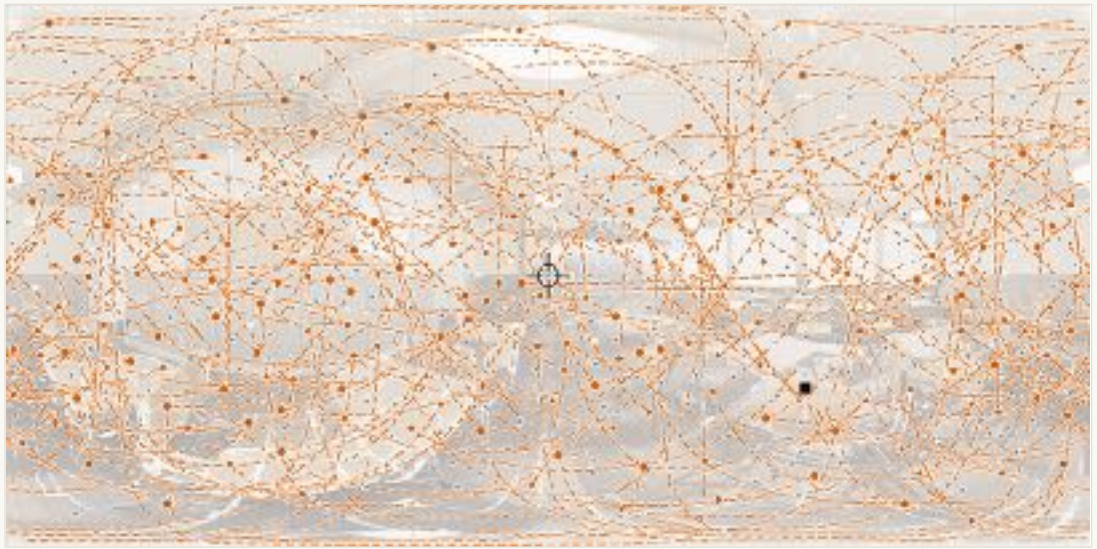
Source: technical report, section 7 (cyclopean coverage memory)

GREEDY FOVEAL EXPLORER V0

Local steps and global jumps



Local: the best of 8 neighbours one step away, scored by edge support × unseen gain.

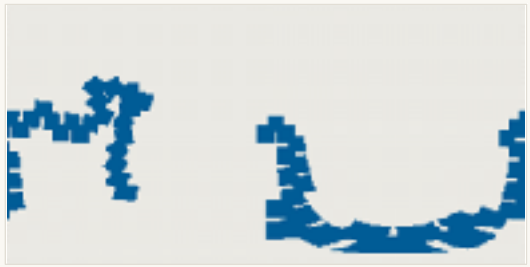


Global: no neighbour qualifies, so jump to the deepest cell of the largest unseen region.

MEASURED · official baseline: 322 local and 219 global saccades; the first global jump is at fixation #69

GREEDY FOVEAL EXPLORER V0

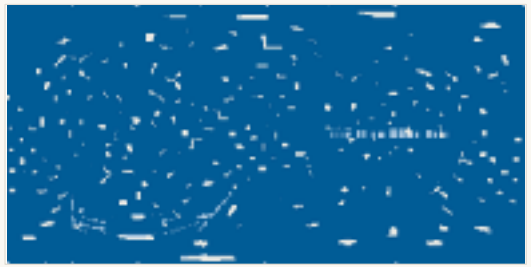
Coverage grows fast, then crawls



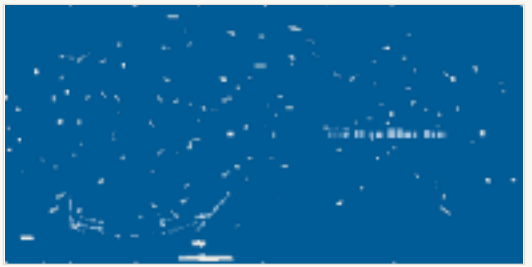
#50 · 14.7 % seen



#200 · 56.1 % seen

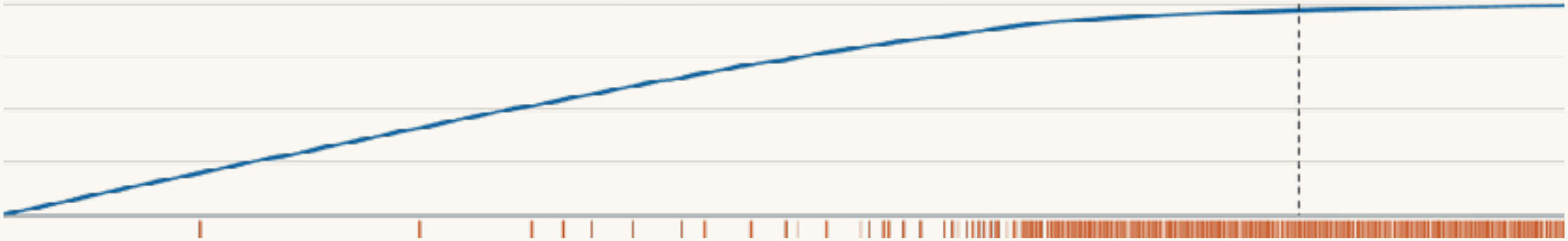


#450 · 96.7 % seen



#542 · 99.0 % seen (stop)

SEEN, % of the sphere, for fixations 1–542 · vermilion ticks: global jumps · dashed line: #450



MEASURED · official baseline run; the last 92 fixations (#451–#542) added only 2.3 points of SEEN

GREEDY FOVEAL EXPLORER V0

One persistent map from 542 fixations

542

fixations: 322 local and 219 global saccades

99.00 %

of the sphere seen; 98.28 % with depth

98.32 %

of the first-hit scene within 12 mm, evaluated only after the run froze

16.5 M

surfels in one head-frame map, built in 733 s



The map from above, ceiling removed

MEASURED · official baseline reproduction (code b77b1d9), bitwise equal to the frozen v0 run in every control-path product

GREEDY FOVEAL EXPLORER V0

The reconstruction against the reference



Reconstructed: the map seen from the head, in its own colours



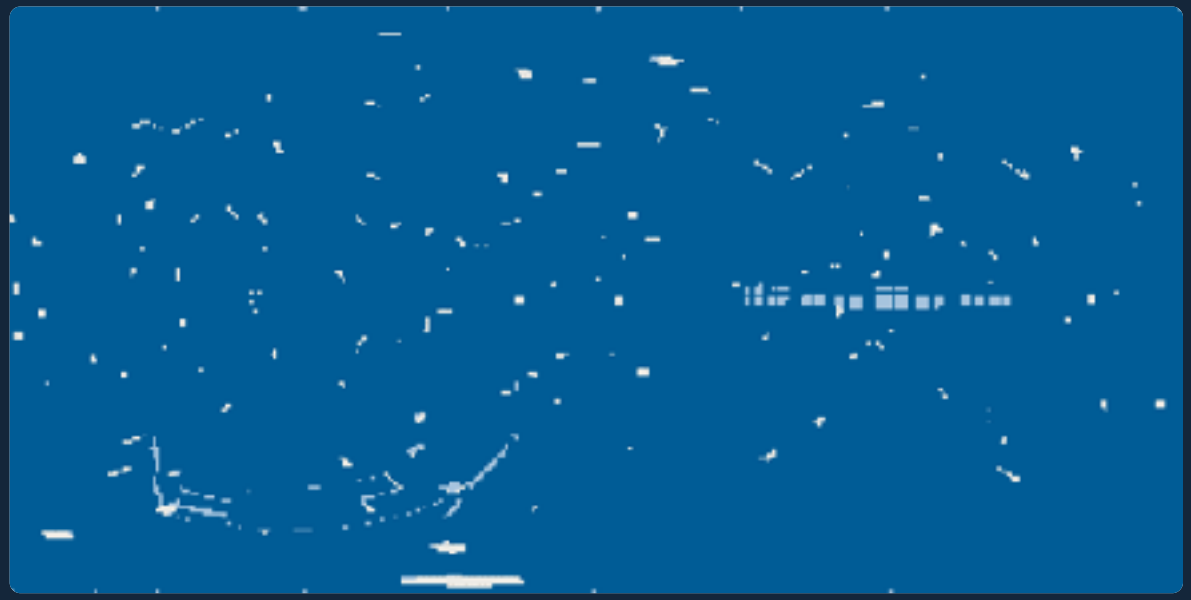
Reference: Blender's first-hit rendering, opened only after the run froze

98.32 % of the first-hit solid angle lies within 12 mm of the map; on covered cells the median distance is 0.73 mm.

MEASURED · official baseline run, post-freeze Breadth-1 evaluation

PLAYGROUND-1 · THE PROBLEM

v0 stops with holes, after a long tail of single-hole jumps



Coverage at the stop, #542: the white specks are UNSEEN holes

184

holes still unseen at the stop (640 cells)

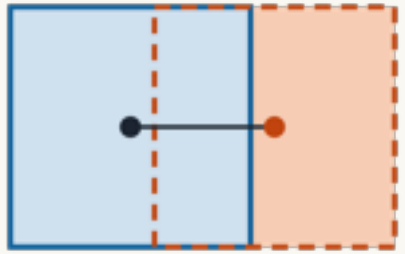
180

fixations in the all-global tail, after the last local step at #362

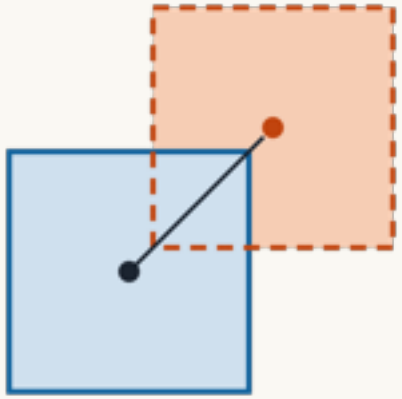
MEASURED · official baseline run (docs/playground/gaze-lattice-report.md, section 4)

PLAYGROUND-1 · THE CAUSE

A diagonal bias leaves slivers behind



Side step: **58 %** of the next footprint is new



Diagonal step: **83 %** new, so the greedy score prefers diagonals

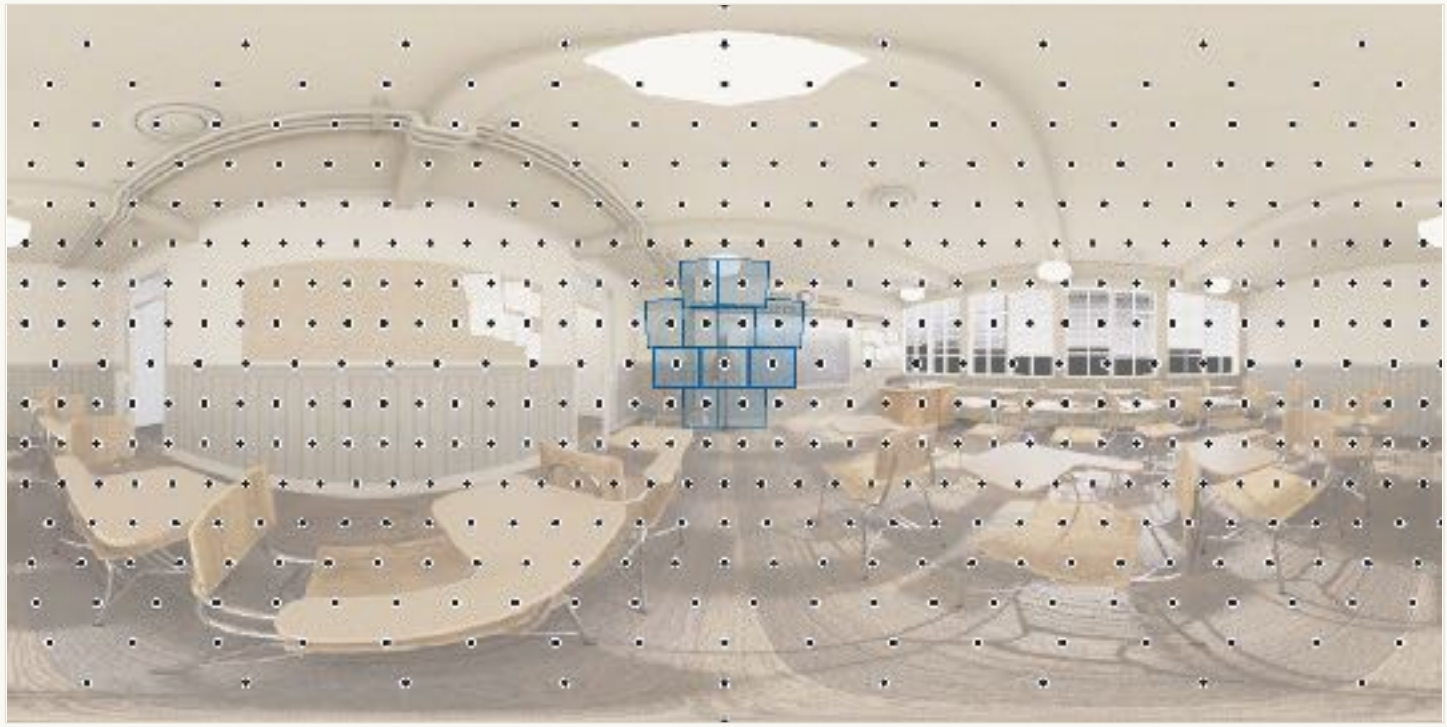
Slivers become the tail

Diagonal staircases, in a tangent frame that drifts with pitch, leave slivers far below 25 % of a footprint. No local move may target them, so each costs one global jump at the end.

Known answer K3 (tools/greedy_foveal/check_lattice.py): first-step unseen gain 0.825 diagonal, 0.58 side

PLAYGROUND-1 · THE FIX

Gaze on a fixed covering lattice



The 496 lattice gazes over the room; outlined, the overlapping 12° footprints at the chalkboard

1. **Lattice**: 10° pitch rings; the 12° squares cover all 64,800 cells.
2. **Local**: any nearby lattice gaze that still adds unseen area.
3. **Global**: the unvisited gaze that adds the most.
4. **Stop**: no lattice gaze adds area, so nothing is unseen.

Contract: docs/playground/gaze-lattice-contract.md · code at tag playground-1-gaze-lattice

PLAYGROUND-1 · THE RESULT

No holes, no tail, fewer fixations

Measure	v0 baseline	Playground-1
Fixations	542	496
Fixations to reach 99 % SEEN	542	472
UNSEEN cells at the stop	640 (184 holes)	0
Global saccades	219	35
All-global tail (fixations)	180	8
Within 12 mm (solid angle)	98.32 %	98.82 %
Wall time	733 s	665 s

Every lattice gaze was fixated exactly once: 496 is the lattice size.

MEASURED · Playground-1 run (code 85fb962) vs the official baseline; docs/playground/gaze-lattice-report.md

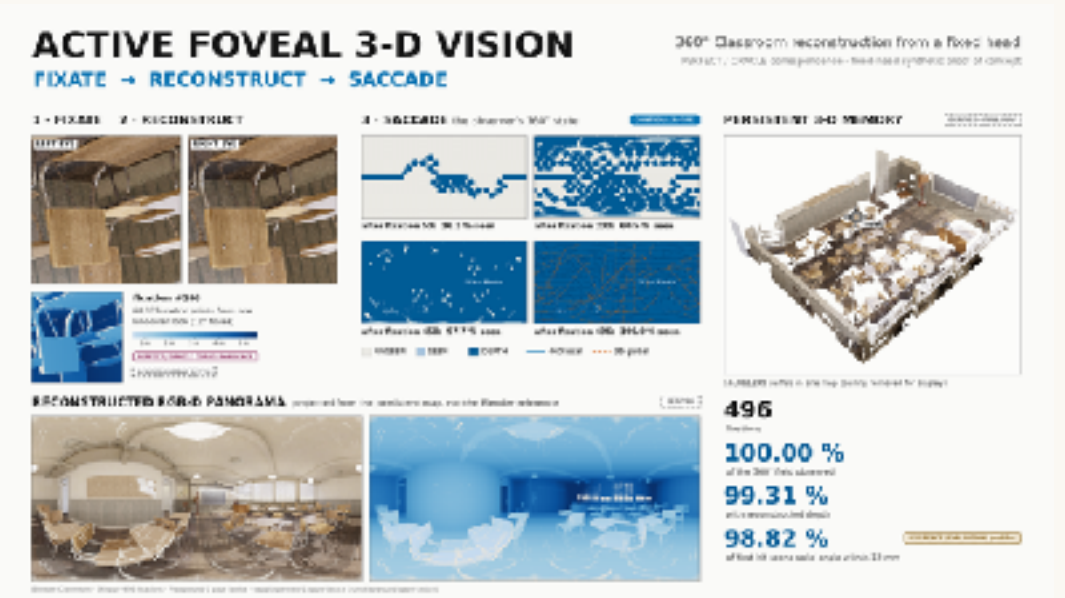
DEMO

Replays of both runs



v0 baseline · 62.4 s · 542 fixations

visuals/greedy-foveal-explorer-v0-demo/demo.mp4



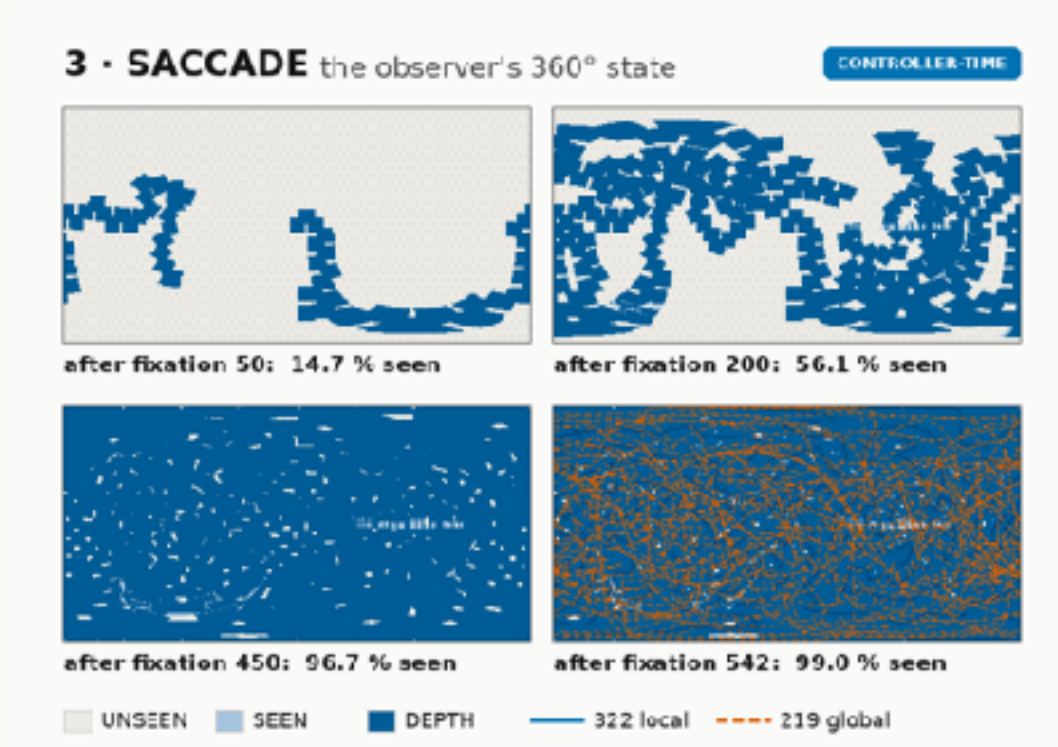
Playground-1 lattice · 53.2 s · 496 fixations

visuals/playground-gaze-lattice-demo/demo.mp4

Offline replays of the frozen runs; presentation only

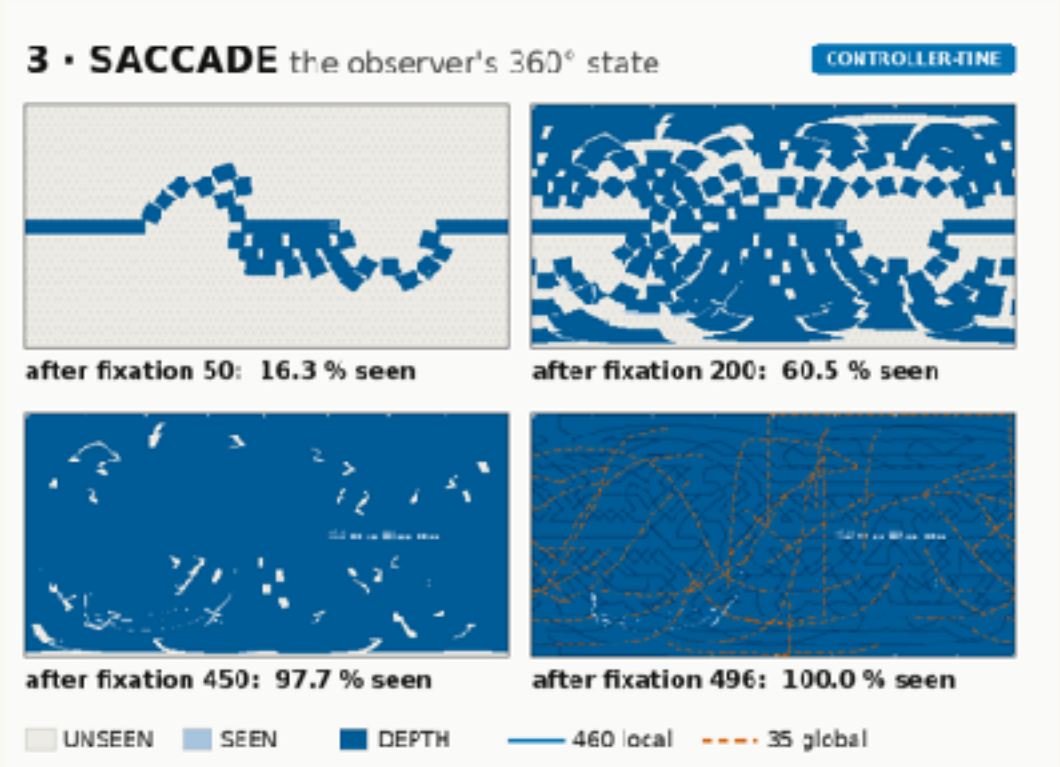
DEMO

Replays of both runs



v0 baseline · 62.4 s · 542 fixations

visuals/greedy-foveal-explorer-v0-demo/demo.mp4



Playground-1 lattice · 53.2 s · 496 fixations

visuals/playground-gaze-lattice-demo/demo.mp4

Offline replays of the frozen runs; presentation only

LIMITS

What this proof of concept does not show

Oracle stereo

Correspondence comes from the renderer, not from images.

Fixed head

Surfaces hidden from one eye stay out of reach of any gaze.

One scene

A static Blender Classroom, rendered at 64 spp.

First hits only

Only surfaces visible from the head are scored.

Storage

About 16.5 M surfels and 1 GB per map.

Lattice size

Playground-1 visits every gaze: 496 is set by the lattice.

NEXT

One component at a time, against the baseline

NEXT EXPERIMENT

Playground-2: a sparser covering lattice

Fewer than 496 gazes, still zero holes; an area bound for 12° squares gives about 290.

Real stereo

Image matching in place of the oracle.

Map-driven

Pick views from 3-D frontiers.

Head motion

Move to see behind occluders.

New scenes

Test beyond the Classroom.